

Jack Cashman

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EDUCATION

University of Rochester

Bachelor of Science, Honors in Computer Science

August 2022 - May 2026

Rochester, NY

Purdue University

Ph.D. in Electrical and Computer Engineering

Starting August 2026

West Lafayette, IN

Relevant Coursework: Advanced Compiler Optimizations, Machine Learning Systems, Computer Organization, Operating Systems, Programming Language Design, Data Structures & Algorithms, Computer Models & Limitations, Honors Research Problem Seminar, Computer Networks.

RESEARCH EXPERIENCE

University of Rochester

Research Assistant Under Prof. Chen Ding

May 2024 – Present

- **Loop Asymptotic Locality Analysis (LALA):** Developed a symbolic tool to model the reuse distance distribution of affine programs, reducing analysis time from doubly exponential to linear relative to program size.
- **Zipfian Caching for Web Workloads:** Designed and evaluated caching policies to outperform LRU on dynamic Zipfian web workloads, achieving a faster convergence to the optimal hit rate.
- **LLM Text Detection:** Used locality measurements of data movement distance for varying N-gram text representation to classify text as human vs. LLM-generated, building an N-gram trace generator for feature extraction.

University of Utah

Research Assistant Under Prof. Ponnuswamy Sadayappan

May 2025 – Present

- Investigated sparse matrix reordering schema cache hierarchy utilization and its effects on Sparse Matrix Vector Multiplication (SpMV) performance.
- Reduced error of a machine learning model for pairwise ranking of data layouts based on predicted L1D, L2, and L3 cache misses, addressing the fragility of traditional reordering heuristics.
- Implemented and validated hardware counter frameworks (**VTune**, **PAPI**, **LIKWID**, **Perf**) to capture precise cache performance data for model training and empirical validation.
- Conducted controlled experiments by disabling hardware prefetching and varying compiler optimization levels to isolate performance effects and align empirical results with model predictions.

Honors Thesis

University of Rochester, Dept. of Computer Science

August 2025 – Present

- Independent research project to develop a novel sparse matrix reordering solution. Approach based on directly leveraging analytical cache modeling to drive reordering decisions.
- Designing an analytical cache model that directly targets performance factors including cache capacity, associativity, latency, and replacement policy, prefetchers, thread count, and more. This method of reordering departs from traditional graph theory approaches.
- Integrating analytical prediction with genetic search and heuristics, validating results with hardware counters.
- Parallelizing reordering search and analytical modeling for improved speed of reordering.

TEACHING EXPERIENCE

Teaching Assistant, Computer Organization
University of Rochester

Aug 2024 – Present

- Led study sessions and office hours to help undergraduate students understand core systems concepts, including memory hierarchy, instruction sets, and digital logic.
- Wrote, graded, and proctored exams.
- Graded student projects and provided feedback on assembly language and C programming assignments.

SELECTED PROJECTS

Tiny-QMOE

Fall 2024

Course Research Project, Machine Learning Systems

- Engineered a quantization and dictionary-based compression pipeline to enable inference of large language models (LLMs) on memory-constrained devices (e.g., mobile phones).
- Achieved **23–35x compression** on LLaMA models with minimal accuracy loss on MMLU and ARC benchmarks.
- Gained hands-on experience with GPU memory hierarchy, low-latency compression algorithms, and LLM benchmarking.

Mid-End Compiler Optimization Framework

Spring 2025

Course Project, Advanced Compilers (Software Analysis & Improvement)

- Built a compiler optimization framework from scratch, operating on an intermediate representation.
- Implemented classic and advanced optimizations including dead code elimination, partial redundancy elimination, and constant propagation using abstract interpretation.

Latency Based Tiling

Spring 2025

Course Research Project, Honors Undergraduate Problem Seminar

- Built a triangle loop based cache latency micro benchmark for approximating cache capacities for L1D, L2, and L3 caches.
- Leveraged the cache approximation for polyhedral program tiling to produce a 21% performance speedup with effectively no compilation overhead unlike auto-tuning.

PUBLICATIONS

- **In Preparation:** Sparse matrix reordering research with Prof. P. Sadayappan.
- **In Preparation:** LALA (Loop Asymptotic Locality Analysis) with Prof. Chen Ding's research group.
- **Preprint:** Jack Cashman. *Latency Based Tiling*. arXiv preprint arXiv:2510.15912, 2025.
- **Preprint:** Jack Cashman, Jiaqi Nie. *Tiny-QMoE*. arXiv preprint arXiv:2509.22951, 2025.

TECHNICAL SKILLS

Languages: Rust, C, RISC-V Assembly, x86-64 Assembly, Coq, CUDA

Developer Tools: Git, Linux/Unix Shell, VTune, PAPI, LIKWID, Perf, GDB

Areas of Interest: Compiler Optimization, High Performance Computing, Locality Theory, Cache Modeling, Computer Architecture, Type Systems/Safety, Machine Verified Proofs, Side Channel Attacks